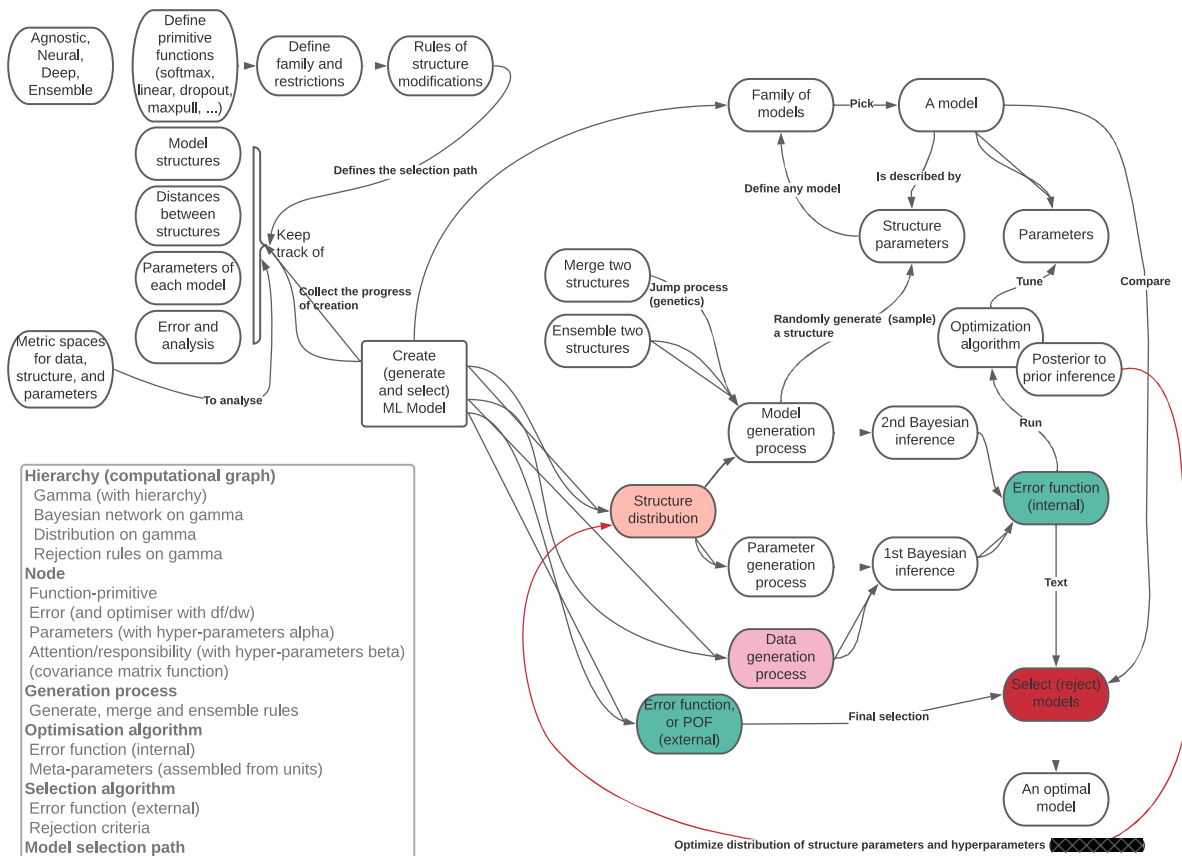


Forward and inverse problems in causal discovery

IA pour la Science

23 April 2026

1 Functional diagram of the model forecasting



2 Generall problem statement in multimodeling

Consider a sample $\mathfrak{D} = \{(\mathbf{x}_i, y_i)\}$, $i \in \mathcal{I} = \overline{1, m}$,
 $\mathbf{x}_i \in \mathbb{R}^n$, $y_i \in \{\pm 1\}$.

- Model weights $\boldsymbol{\pi}$ in mixture come from Dirichlet distribution, i.e.
 $p(\boldsymbol{\pi}|\mu) = \text{Dir}(\boldsymbol{\pi}|\mu)$
- Models' parameters $\mathbf{w}_k$ are sampled from normal distribution, i.e.
 $p(\mathbf{w}_k|\mathbf{A}_k) = \mathcal{N}(\mathbf{w}_k|\mathbf{0}, \mathbf{A}_k^{-1})$, $k = 1, \dots, K$.
- For each object $\mathbf{x}_i$ a model describing f_{k_i} describing it is selected independently, so that $p(k_i = k) = \pi_k$.
- For each object $\mathbf{x}_i$ class label y_i is defined according to the model f_{k_i} :
 $y_i \sim \text{Be}(\sigma(\mathbf{w}_{k_i}^\top \mathbf{x}_i))$.

Call a multimodel **optimal**, iff it has the maximum evidence

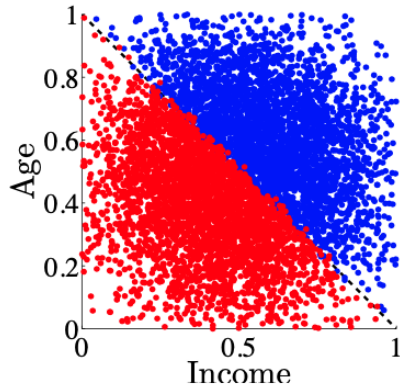
$$[\mathbf{A}_1^*, \dots, \mathbf{A}_K^*, (\mu^*)] = \arg \max_{\mathbf{A}_1, \dots, \mathbf{A}_K, (\mu)} p(\mathbf{y}|\mathbf{X}, \mathbf{A}_1, \dots, \mathbf{A}_K, (\mu)),$$

where $\mathbf{A}_1 \in Q_{\mathbf{A}_1}$, $\dots$, $\mathbf{A}_K \in Q_{\mathbf{A}_K}$, $(\mu \in Q_\mu)$.

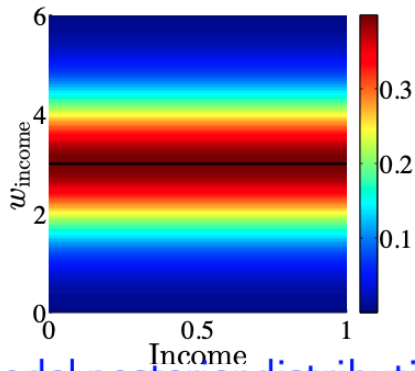
The goal is to construct the optimal trained multimodel

$$[\mathbf{w}_1^*, \dots, \mathbf{w}_K^*, (\boldsymbol{\pi}^*)] = \arg \max_{\mathbf{w}_1, \dots, \mathbf{w}_K, (\boldsymbol{\pi})} p(\mathbf{w}_1, \dots, \mathbf{w}_K, (\boldsymbol{\pi})|\mathbf{X}, \mathbf{y}, \mathbf{A}_1^*, \dots, \mathbf{A}_K^*, (\mu^*))$$

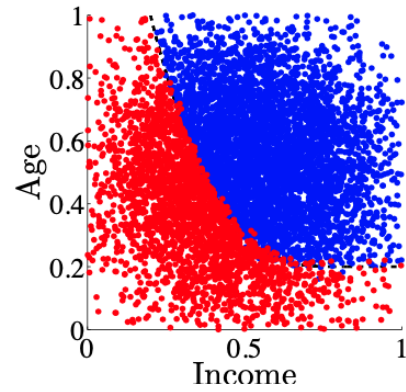
3 When do we need a multimodel



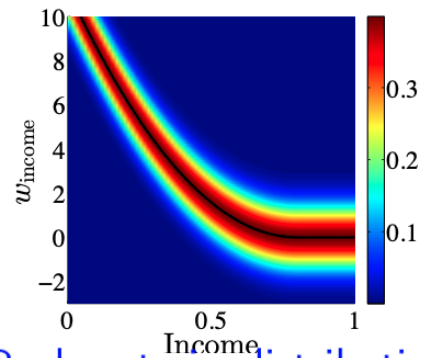
Sample generation assumption



Model posterior distribution
 $p(\mathbf{w} | \mathbf{X}, \mathbf{y})$



Real data



Real posterior distribution
 $p(\mathbf{w} | \mathbf{X}, \mathbf{y})$

4 Model comparison hypotheses

Call a multimodel with joint distribution

$p(\mathbf{y}, \mathbf{w}_1, \dots, \mathbf{w}_K, (\boldsymbol{\pi}) | \mathbf{X}, \mathbf{A}_1, \dots, \mathbf{A}_K, (\mu))$ (s, α) -adequate, if the models, which constitute the multimodel are pairwise statistically distinguishable with similarity function s and significance level α .

Issue

Despite multimodel pruning due to Dirichlet prior, it can still be not (s, α) – adequate, i.e. can contain similar models.

Consider

- Two models f_1 and f_2 , with parameters $\mathbf{w}_1, \mathbf{w}_2$.
- Samples $(\mathbf{X}_1, \mathbf{y}_1)$ and $(\mathbf{X}_2, \mathbf{y}_2)$,
 $y_{1,i} = f_1(\mathbf{x}_{1,i}, \mathbf{w}_1), y_{2,i} = f_2(\mathbf{x}_{2,i}, \mathbf{w}_2)$.
- Prior distributions $\mathbf{w}_1 \sim p_1(\mathbf{w}), \mathbf{w}_2 \sim p_2(\mathbf{w})$.
- Posterior distributions $g_1(\mathbf{w}_1) = p(\mathbf{w}_1 | \mathbf{X}_1, \mathbf{y}_1)$ and $g_2(\mathbf{w}_2) = p(\mathbf{w}_2 | \mathbf{X}_2, \mathbf{y}_2)$.

Goal: to construct a similarity function on a pair of distributions $g_1(\mathbf{w})$ and $g_2(\mathbf{w})$, satisfying a list of criteria.

5 Proposed statement of the model comparison

Generalized linear model with natural link function and parameters' prior distribution $p(\mathbf{w}|\mathbf{A})$ is a probabilistic model with joint distribution $p(\mathbf{y}, \mathbf{w}|\mathbf{X}, \mathbf{A}) = p(\mathbf{y}|\mathbf{X}, \mathbf{w})p(\mathbf{w}|\mathbf{A})$,

where $p(\mathbf{y}|\mathbf{X}, \mathbf{w}) = \prod_{i=1}^m p(y_i|\mathbf{x}_i, \mathbf{w})$,

$p(y_i|\mathbf{x}_i, \mathbf{w}) = c(y_i, \tau) \exp((\theta_i y_i - b(\theta_i))/d(\tau))$, where $\theta_i = \mathbf{w}^\top \mathbf{x}_i$.

Theorem

Consider $g_1 = \mathcal{N}(\mathbf{v}_1, \mathbf{\Sigma}_1)$, $g_2 = \mathcal{N}(\mathbf{v}_2, \mathbf{\Sigma}_2)$.

Then for $s(g_1, g_2)$ we get

$$s(g_1, g_2) = \exp \left(-\frac{1}{2}(\mathbf{v}_1 - \mathbf{v}_2)^\top (\mathbf{\Sigma}_1 + \mathbf{\Sigma}_2)^{-1}(\mathbf{v}_1 - \mathbf{v}_2) \right).$$

6 Explicit form of $s(g_1, g_2)$ for a pair of normal distributions

Denote pairwise similarity matrix by $\mathbf{S} = [s_{kl}(g_k(\mathbf{w}_k), g_l(\mathbf{w}_l))]$, and a corresponding matrix of reached significance level by

$$\mathbf{T} = [\mathbf{P}(s(g_k(\mathbf{w}_k), g_l(\mathbf{w}_l)) < s_{kl} | \mathbf{w}_k = \mathbf{w}_l)], \quad k, l = 1, \dots, K.$$

1 Find $[k^*, l^*] = \arg \max_{k < l} t_{kl}$.

2 If $t_{k^*l^*} < \alpha$, stop, else go to step 3.

3 ■ *For multilevel models:*

Unite models k^*, l^* and recalculate posterior $g_{k^*}(\mathbf{w}_{k^*})$.

$$\mathcal{I}_{k^*} \sqcup \mathcal{I}_{l^*} \rightarrow \mathcal{I}_{k^*}, \quad \mathbf{A}_{k^*}^* = \arg \max_{\mathbf{A}_{k^*}^*} p(\mathbf{y}_{\mathcal{I}_{k^*}} | \mathbf{X}_{\mathcal{I}_{k^*}}, \mathbf{A}_{k^*}^*);$$

$$g_{k^*}(\mathbf{w}_{k^*}) = \mathcal{N}(\mathbf{w}_{k^*} | \mathbf{w}_{k^*}^*, \Sigma_{k^*}^*)$$

■ *For mixture of models:*

Unite models k^*, l^* and reoptimize the mixture. New initial point:

$$\pi_{k^*} + \pi_{l^*} \rightarrow \pi_{k^*}, 0 \rightarrow \pi_{l^*}, \frac{\mathbf{w}_{k^*} + \mathbf{w}_{l^*}}{2} \rightarrow \mathbf{w}_{k^*}, \mathbf{w}_k \rightarrow \mathbf{w}_k, k \neq k^*, l^*.$$

4 Delete column l^* from matrices $\mathbf{S}$ and $\mathbf{T}$, and recalculate s_{k^*l}, t_{k^*l} for $l \neq k^*$. Go to step 1.

7 Mixture of experts as a multimodel

The gate function $\pi_k(\mathbf{x}) : \mathbf{x} \rightarrow [0; 1]$ defines evidence if k -th model at $\mathbf{x}$.

$$\pi_k(\mathbf{x}, \mathbf{V}) = \sigma\left(\mathbf{g}(\mathbf{x}, \omega), \mathbf{V}\right) = \frac{\exp \mathbf{v}_k^\top \mathbf{g}(\mathbf{x}, \omega)}{\sum_{k'=1}^K \exp \mathbf{v}_{k'}^\top \mathbf{g}(\mathbf{x}, \omega)},$$

where $\mathbf{V} = [\mathbf{v}_1, \dots, \mathbf{v}_K, \omega]$, σ is softmax,

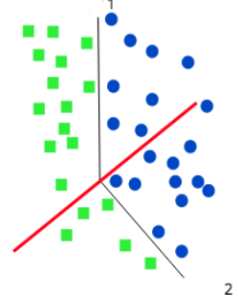
$\mathbf{g}(\mathbf{x}, \omega)$ — an expert, a model at $\mathbf{x}$.

$$\text{Mixture of experts: } \mathbf{f}_{\text{me}}(\mathbf{x}) = \sum_{k=1}^K \pi_k(\mathbf{x}, \mathbf{V}) f_k(\mathbf{x}, \mathbf{w}).$$

The posterior for $\mathbf{y}$ is

$$\begin{aligned} p(\mathbf{y}|\mathbf{x}, \theta) &= \sum_{k=1}^K p(k|\mathbf{x}, \theta) p(\mathbf{y}|k, \mathbf{x}, \theta) = \\ &= \sum_{k=1}^K \pi_k(\mathbf{x}, \mathbf{V}) \exp\left(-\frac{1}{2\beta_k} (\mathbf{y} - f_k(\mathbf{x}, \mathbf{w}))^2\right), \text{ where} \end{aligned}$$

$$\mathbf{y} \sim \mathcal{N}(\mathbf{f}_k(\mathbf{x}, \mathbf{w}_k), \beta_k), \quad \theta = [\mathbf{w}_1, \dots, \mathbf{w}_K, \mathbf{V}, \beta].$$



8 Similarity of models in a multimodel

Call a multimodel with joint distribution

$p(\mathbf{y}, \mathbf{w}_1, \dots, \mathbf{w}_K, (\boldsymbol{\pi}) | \mathbf{X}, \mathbf{A}_1, \dots, \mathbf{A}_K, (\mu))$ (s, α) -adequate, if the models, which constitute the multimodel are pairwise statistically distinguishable with similarity function s and significance level α .

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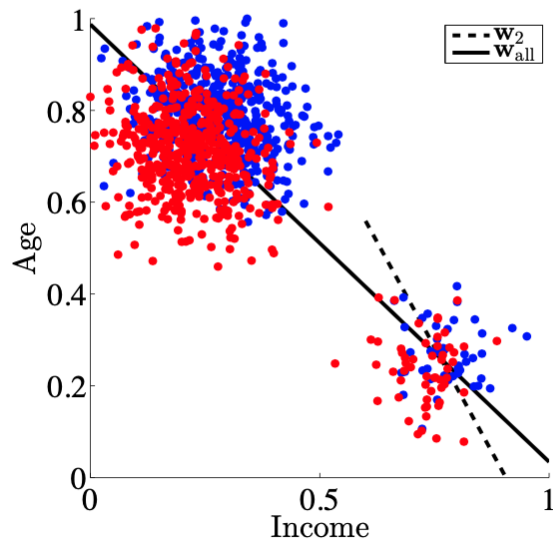
Consider

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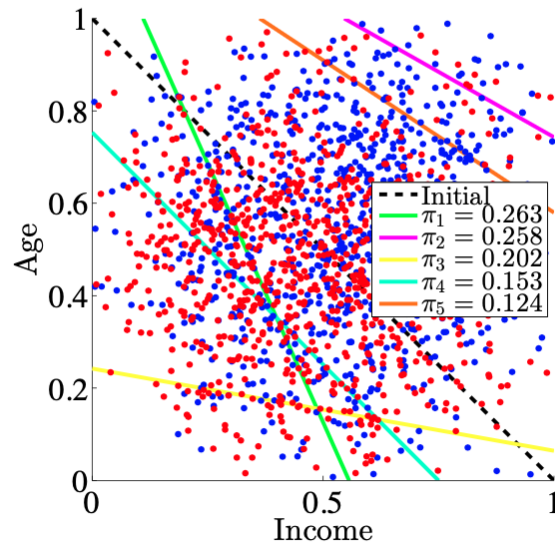
Goal: to construct a similarity function on a pair of distributions $g_1(\mathbf{w})$ and $g_2(\mathbf{w})$, satisfying a list of criteria.

9 Model comparison issues

Issue: large number of close or identical models lead to loss of interpretability and poor forecast quality.



Inadequate multilevel model



Inadequate mixture of models

10 Requirements to the similarity function

Correct similarity function s

- 1 is defined for unequal support distributions,
- 2 $s(g_1, g_2) \leq s(g_1, g_1)$,
- 3 $s \in [0, 1]$,
- 4 $s(g_1, g_1) = 1$,
- 5 is close to 1, if $g_2(\mathbf{w})$ is uninformative distribution,
- 6 symmetric, $s(g_1, g_2) = s(g_2, g_1)$.

Theorem

Similarity functions generated by Kullback-Leibler and Jensen-Shannon divergence, Hellinger and Bhattacharyya distances are not correct.